



Jordan University of Science and Technology
Faculty of Science and Arts
Department of Biotechnology and Genetic Engineering
Microbial Biotechnology Lab. (BT 431)
Experiment # 3: Isolation of Coliforms and Fecal Coliforms

Microscopic characters	<i>E. coli</i>
Gram reaction	- ve
Shape	Coccobacilli
Arrangement	Single
Spore formation	Non-spore former

Colony morphology		<i>E. coli</i>
	Size	Large
	Surface	Smooth
	Pigmentation	Grayish white
Forms	Configuration	Round
	Margins	Entire margins
	Elevation	Low convex

Medium:	Lauryl Sulphate Broth (Lauryl Tryptose Broth)	
Purpose:	Used for the detection of coliforms* in water, wastewater, dairy products and other food samples.	
Ingredient	Function	
Tryptose	Provides essential growth substances, such as nitrogen and carbon compounds, sulphate and trace ingredients	
Lactose	The fermentable carbohydrate	
Sodium chloride	Maintains osmotic equilibrium	
Dipotassium phosphate	Provide buffering system	
Monopotassium phosphate		
Sodium lauryl sulphate	Inhibits organisms other than coliforms	
Notes		
Incubated with inverted Durham’s tube	To detect the production of gas	
Results		
Bacteria	Appearance	Reason
<i>Escherichia coli</i>	Turbidity with	Fecal coliform
	gas	Fermentations yields acid + gas
<i>Salmonella typhimurium</i>	Good to excellent growth with no gas	
<i>Staphylococcus aureus</i>	Inhibited	Not a coliform

* Coliforms grown in this medium ferment lactose producing gas while other bacteria are either inhibited or grow without gas production

Medium:	EC Broth	
Purpose:	Used for the selective enumeration of presumptive <i>Escherichia coli</i> by MPN technique from water samples and from clinical samples.	
Ingredient	Function	
Tryptone	Provides nitrogenous and carbonaceous compounds, long chain amino acids and other essential growth nutrients	
Lactose	The fermentable sugar	
Bile salts mixture	Inhibit gram-positive bacteria especially bacilli and fecal Streptococci	
Dipotassium phosphate	Form the buffering system	
Monopotassium phosphate		
Sodium chloride	Maintains osmotic equilibrium	
Notes		
Incubated with inverted Durham's tube	To detect the production of gas	
Incubated at 44.5 - 45.5 ° C	For detection of fecal coliform (<i>E. coli</i>) *	
Results		
Bacteria	Appearance	Reason
<i>Escherichia coli</i>	Turbidity with	Fecal coliform
	gas	Fermentation yields acid + gas

* Fecal (thermotolerant) coliforms (e.g *E. coli*) are able to grow at such a high temperature

Medium:	Brilliant Green Bile Broth 2%	
Purpose:	Used for the detection and confirmation of coliform bacteria in water, wastewater, foods, milk and dairy products	
<i>Ingredient</i>	<i>Function</i>	
Peptone	A source of essential nutrients	
Lactose	The fermentable carbohydrate	
Bile	Inhibits gram-positive bacteria	
Brilliant green	Inhibits gram-negative bacteria	
<i>Notes</i>		
Incubated with inverted Durham’s tube	To detect the production of gas *	
<i>Results</i>		
Bacteria	Appearance	Reason
<i>Escherichia coli</i>	Turbidity with	Fecal coliform
	gas	Fermentation yields acid + gas
<i>Bacillus cereus</i>	Inhibited	Gram + ve

* Production of gas indicates the positive evidence of faecal coliform since non-faecal coliforms growing in this medium do not produce gas.

Medium:	EMB Agar	
Purpose:	Used for the isolation and differentiation of gram-negative enteric bacteria from clinical and nonclinical specimens	
<i>Ingredient</i>	<i>Function</i>	
Peptic digest of animal tissue	Source of carbon, nitrogen, and other essential growth nutrients	
Dipotassium phosphate	Forms the buffering system	
Lactose	Sources of fermentable carbohydrates	
Sucrose		
Eosin - Y	Serve as differential indicators	
Methylene blue		
Agar	Solidifying agent	
<i>Results</i>		
Bacteria	Appearance	Reason
<i>Klebsiella pneumoniae</i>	Pink, mucoid glistening colonies	Gram -ve, Lactose fermenter
<i>Salmonella Typhimurium</i>	Luxuriant growth, colorless colonies	Gram -ve, Lactose non-fermenter
<i>Staphylococcus aureus</i>	Inhibited	Gram +ve
<i>Escherichia coli</i>	Dark purple with black center and green metallic sheen	Gram -ve, Lactose fermenter



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Experiment # 4: Isolation and Identification of *Salmonella*

Microscopic characters	
Gram reaction	- ve
Shape	Coccobacilli
Arrangement	Single
Spore formation	Non-spore former

Colony morphology		Salmonella
Size		Medium
Surface		Smooth
Pigmentation		Grayish white
Forms	Configuration	Round
	Margins	Entire margins
	Elevation	Convex

Medium:	Selenite Cystine Broth	
Purpose:	Used as selective enrichment media for <i>Salmonella</i>	
Ingredient	Function	
Tryptone	Provides nitrogenous and carbonaceous compounds, long chain amino acids, vitamins and other essential nutrients	
Lactose	Maintains the pH of medium	
Disodium phosphate	Maintains a stable pH and also lessens the toxicity of selenite	
L-Cystine	Source of amino acid and improves recovery of <i>Salmonella</i>	
Sodium selenite	Inhibits Gram-positive bacteria and most enteric Gram-negative bacteria except <i>Salmonella</i>	
Results		
Bacteria	Appearance	Reason
<i>Escherichia coli</i>	None - poor	enteric Gram-negative
<i>Salmonella Typhi</i>	good - luxuriant	

Medium:	Xylose-Lysine Deoxycholate Agar (XLD Agar)	
Purpose:	is a selective medium recommended for the isolation and enumeration of <i>Salmonella Typhi</i> and other <i>Salmonella</i> species.	
<i>Ingredient</i>	<i>Function</i>	
Yeast extract	Provide nitrogen and vitamins	
L-Lysine	Source of amino acids	
Lactose	Sources of fermentable carbohydrates	
Sucrose		
Xylose		
Sodium chloride	Maintains the osmotic balance of the medium	
Sodium deoxycholate	Selective agent that inhibits the growth of gram-positive microorganisms	
Sodium thiosulphate	Sulfur source	
Ferric ammonium citrate	Sulfur indicator	
Phenol red	Ph indicator	
Agar	Solidifying agent	
<i>Results</i>		
Bacteria	Appearance	Reason
<i>Salmonella Typhi</i>	luxuriant	Gram -ve
	red colonies with black centers	Xylose fermenter, lysine decarboxylase +ve & H ₂ S producer
<i>Salmonella Paratyphi A</i>	good - luxuriant	Gram -ve
	red colonies	Xylose fermenter, lysine decarboxylase +ve & H ₂ S non-producer
<i>Staphylococcus aureus</i>	Inhibited	Gram +ve
<i>Escherichia coli</i>	Fair	Gram -ve
	yellow colonies	Lactose fermenter
<i>Shigella dysenteriae</i>	good-luxuriant	Gram -ve
	red colonies	Xylose non-fermenter, lysine decarboxylase -ve & H ₂ S non-producer

Medium:	Bismuth Sulphite Agar
Purpose:	Used a selective medium recommended for the isolation and identification of <i>Salmonella Typhi</i> and other <i>Salmonella</i> species.
<i>Ingredient</i>	<i>Function</i>
Peptone	Source as carbon, nitrogen, long chain amino acids, vitamins and essential growth factors
Dextrose (Glucose)	Sources of fermentable carbohydrates
Disodium phosphate	Maintains the osmotic equilibrium
Ferrous sulphate	Aids in detection of hydrogen sulphide production
Bismuth sulphite indicator	Inhibit the intestinal gram-positive and gram-negative bacteria
Brilliant green	
Agar	Solidifying agent
<i>Results</i>	
Bacteria	Appearance
<i>Salmonella Typhi</i>	luxuriant
	Black colonies with metallic sheen



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Experiment # 5: Isolation and Identification of *Staphylococcus aureus*

Microscopic characters	
Gram reaction	+ ve
Shape	Cocci
Arrangement	Grape-like clusters
Spore formation	Non-spore former

Colony morphology		<i>Staphylococcus aureus</i>
Size		Large
Surface		Smooth
Pigmentation		Golden-yellow
Forms	Configuration	Round
	Margins	Smooth
	Elevation	Raised

Medium:	Mannitol Salt Agar (MSA)
Purpose:	A <u>selective differential</u> medium for the isolation of presumptive pathogenic staphylococci. Most other bacteria are inhibited, with the exception of a few halophilic species
<i>Ingredient</i>	<i>Function</i>
Proteose peptone	Provide nitrogen source
Beef extract	
Sodium chloride	Serves as an inhibitory agent against bacteria other than staphylococci
D-Mannitol	The fermentable carbohydrate
Phenol red	pH indicator
Agar	Solidifying agent
<i>Additives</i>	
Egg yolk emulsion	Helps in differentiation of lipase activity of <i>Staphylococcus aureus</i> which can be visualized as yellow opaque zones around the colonies

Results		
Bacteria	Appearance	Reason
<i>Staphylococcus epidermidis</i>	fair - good	Gram +ve
	red colonies	Mannitol non-fermenter
	red zone around the colony	
<i>Escherichia coli</i>	Inhibited	Gram -ve non-halophilic
<i>Staphylococcus aureus</i>	luxuriant	Gram +ve, halophilic
	yellow colonies	Mannitol fermenter
	yellow zone around the colony	

Medium:	Toluidine Blue DNA Agar	
Purpose:	Used for detection of thermostable deoxyribonuclease activity to establish speciation of <i>S.aureus</i> in contaminated foods.	
<i>Ingredient</i>	<i>Function</i>	
Deoxyribonucleic acid (DNA)*	Substrate of DNase	
Calcium chloride *	Provide the ions and maintain osmotic equilibrium	
Sodium chloride		
Toluidine blue	Aids in detection of DNase activity by the production of a visible red (pink) colored reaction	
Tris (hydroxymethyl) amino methane	Forms the buffering system	
Agar	Solidifying agent	
<i>Results</i>		
Bacteria	Appearance	Reason
<i>Staphylococcus aureus</i>	Red (pink) zones	DNase +ve
<i>Staphylococcus epidermidis</i>	No zones	DNase -ve

* The addition of DNA & Calcium chloride activates the enzyme DNase.

Medium:	Brain Heart Infusion Broth
Purpose:	For propagation of pathogenic cocci and other fastidious organisms
Ingredient	Function
Calf brain infusion	Sources of carbon, nitrogen, essential growth factors, amino acids and vitamins
Beef heart infusion	
Proteose peptone	
Dextrose	The fermentable carbohydrate (D-glucose)
Sodium chloride	The osmotic stabilizer
Disodium phosphate	Helps in maintaining the buffering action of the medium

Characteristic	<i>S. aureus</i>	<i>S. epidermidis</i>	Micrococci
Catalase activity	+	+	+
Coagulase production	+	-	-
Thermonuclease production	+	-	-
Lysostaphin* sensitivity	+	+	-
Anaerobic utilization of:			
Glucose	+	+	-
Mannitol	+	-	-

* Lysostaphin: is an extracellular enzyme secreted by strains of *S. simulans* biovar *staphylolyticus* with a bactericidal activity against staphylococci based on its capability of cleaving the peptidoglycan present in the bacterial cell walls.



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Experiment # 6: Isolation and Identification of *Bacillus cereus*

Microscopic characters		
Gram reaction	+ ve	
Shape	Rods (square ends)	
Arrangement	Chains	
Spore formation	Central / subcentral without swelling	

Colony morphology		<i>Bacillus cereus</i>	
Size		Large	
Surface		Rough	
Pigmentation		Grey-white	
Forms	Configuration	Irregular	
	Margins	Less wavy edge	
	Elevation	Raised	

Medium:	MYP Agar (Phenol Red Egg Yolk Polymyxin Agar)
Purpose:	Selective and differential medium used for isolation and identification of pathogenic Staphylococci and Bacillus species
<i>Ingredient</i>	<i>Function</i>
Peptic digest of animal tissue	Provide nitrogen source
Meat extract	
D-Mannitol	The fermentable carbohydrate
Sodium chloride	The osmotic stabilizer
Phenol red	Ph indicator
Agar	Solidifying agent
<i>Additives</i>	
Egg yolk emulsion	A source of lecithin to help in differentiation of lecithinase producing colonies
Polymyxin B Sulphate	Helps to restrict growth of gram-negative bacteria such as <i>Escherichia coli</i> and <i>Pseudomonas aeruginosa</i>

Results		
Bacteria	Appearance	Reason
<i>Bacillus cereus</i>	luxuriant	Gram +ve, spore former
	red colonies	mannitol non-fermenter
	opaque zone around the colony	Lecithinase positive
<i>Bacillus subtilis</i>	luxuriant	Gram +ve, spore former
	yellow colonies	mannitol fermenter
	no zone around the colony	Lecithinase negative
<i>Escherichia coli</i>	No growth	Gram -ve
<i>Pseudomonas aeruginosa</i>	No growth	Gram -ve
<i>Staphylococcus aureus</i>	luxuriant	Gram +ve
	yellow colonies	mannitol fermenter
	opaque zone around the colony	Lecithinase positive

Medium:	Phenol Red Dextrose Broth		
Purpose:	Used for dextrose fermentation studies of microorganisms.		
Ingredient	Function		
Proteose peptone	Provide nitrogen source		
Beef extract			
Sodium chloride	The osmotic stabilizer (maintains the osmotic equilibrium of the medium)		
Dextrose	The fermentable carbohydrate (D-glucose)		
Phenol red	Ph indicator		
Results			
Bacteria	Appearance		Reason
<i>Bacillus cereus</i>	Growth	luxuriant	
	Acid	Medium turns yellow	Glucose fermentation yields acid
	Gas	No gas	Glucose fermentation yields acid but not gas
<i>Escherichia coli</i>	Growth	luxuriant	
	Acid	Medium turns yellow	Glucose fermenter
	Gas	Positive	Glucose fermentation yields acid with gas
<i>Salmonella Typhi</i>	Growth	luxuriant	
	Acid	Medium turns yellow	Glucose fermentation yields acid
	Gas	Positive	Glucose fermentation yields gas
<i>Shigella flexneri</i>	Growth	luxuriant	
	Acid	Medium turns yellow	Glucose fermentation yields acid
	Gas	No gas	Glucose fermentation yields acid but not gas

Medium:	Nutrient Agar w/ Tyrosine	
Purpose:	Used for cultivation and enumeration of <i>Bacillus cereus</i> in water and food	
Ingredient	Function	
Beef extract	Provide essential nutrients for the growth of the organism.	
Peptone		
Agar	Solidifying agent	
Tyrosine	Source of amino acid	
Results		
Bacteria	Appearance	Reason
<i>Bacillus cereus</i>	good-luxuriant growth	Able to hydrolyze tyrosine
	Clearing of medium in and around the bacterial growth (positive reaction)	
<i>Escherichia coli</i>	good	Unable to hydrolyze tyrosine
	No clear zones (negative reaction)	

Test	Medium	Result	Conclusion
MR-VP tests	MR-VP broth	MR: The medium shows no change to red	MR negative
		VP: Red ring forms on the surface of the medium	VP positive



MICROBIAL BIOTECHNOLOGY LAB



**BIOTECHNOLOGY & GENETIC
ENGINEERING**

MIDTERM EXAM

**LAB PROFESSOR:
ZIAD JARADAT**

**LAB INSTRUCTOR:
MOHAMAD SHATNAWI**



Referance:

- Lab notes
- Lab Manual

**DONE BY :
JULNAR AYMAN**

LAB#1(FOOD MICROBIOLOGY)

Food microbiology :the study of the microorganism that inhibit, create, or contaminate food.

★ **Create:** cheese ,yoghurt

★ **Contaminate:**

1.Gram positive (+ve):

Bacilli: bacillus cereus

Cocci: staphylococcus

*Staphylococcus:
non-sporeformer

2.Gram negative(-ve):

salmonella

Coli form :E-coli



Method used to isolate bacteria:

1.isolation: serial dilution by (surface spreading, pour plate)

Result: isolate colonies/contaminated, not contaminated.

2. purification: by (surface streaking)

(eg:quadrant method)

Result: pure culture

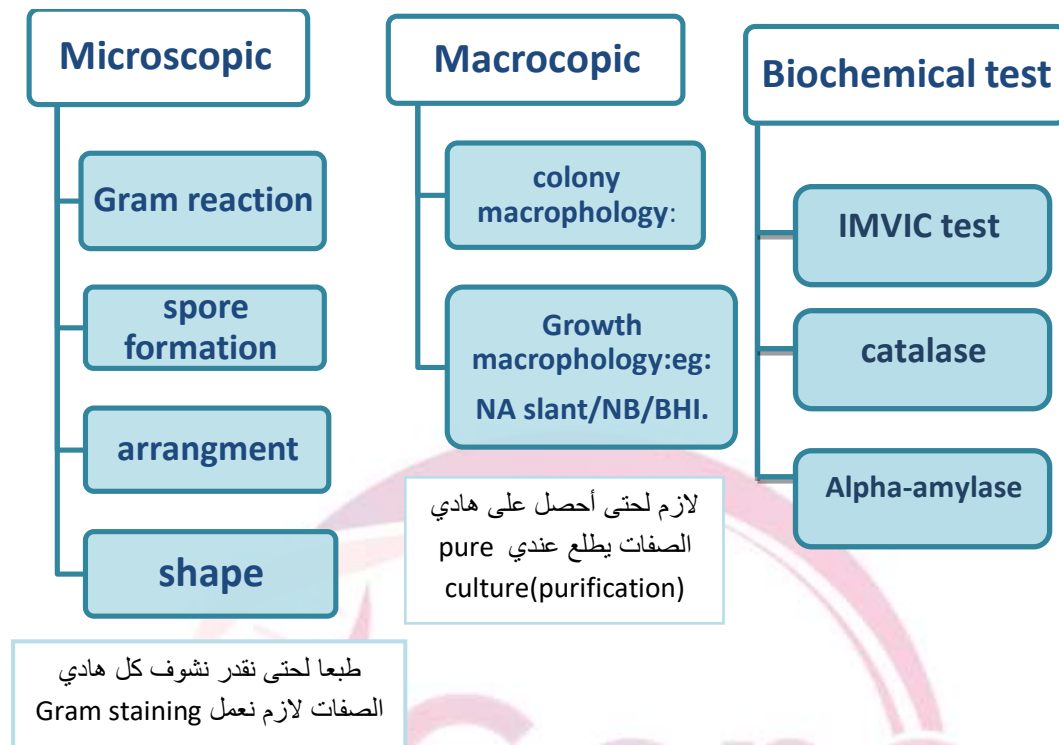
ال(contaminated: عدة أنواع من البكتيريا
بس من مصدر واحد /

ال not contaminated: عدة أنواع من
البكتيريا ومن أكثر من مصدر)

الpure culture: بحتوي على نوع واحد من
البكتيريا

أستاذ محمد رح ينهكم انه
ال quadrant method
هي أفضل طريقة تحديدا
للطلاب

3.identification:



★ NA slant : Nutrient agar slant



★ BHI :brain heart infusion agar



★ NB: Nutrient broth(is the nutrient agar lack of the solidifying agent)



LAB#2(Sampling & Handling the sample)

During sampling, you take the following in considerate:

1.Date

2.collection:a.presentative:eg: separate unite(sample must equal).

$$\sqrt{\text{totalsample}} \leq 10 \text{ units}$$

b. condition of the sample c. Aseptic condition

3.Handling

4.Analysis:all sample analysis as follow:

➡ a. Blending:

if sample need blending:

ليش أنا كاتبالكم؟ if sample need blending?

لانه نحنا ممكن نشغل على sample ما تحتاج blending مثل الحليب لأنه أصلا سائل وما في اي قطع تاتر خلال ال seeding /يعني اذا الاستاذ طلب منكم خطوات ال analyzing a milk sample بتستغنوا عن هادي الخطوة وبتكتبوا add direct to tube

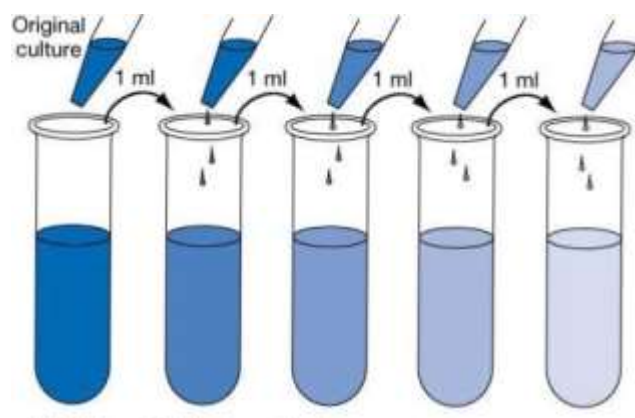
weight sample ➡ sterile blender ➡ H2O ➡ blend

if blending not required, sample added to the dilution blank

Blend: 2 min high speed

/2 min low speed

➡ b. Dilution:



C. Seeding: inoculation, use specific media (eg: Nutrient agar)

#LAB WORK:

We will use fresh material(eg:tomato)



*each 11g of fresh material =1g dry wt=1cm³=1ml

نوت: في groups رح يشتغلوا
على /milk يعني مو شرط بس
بندورة

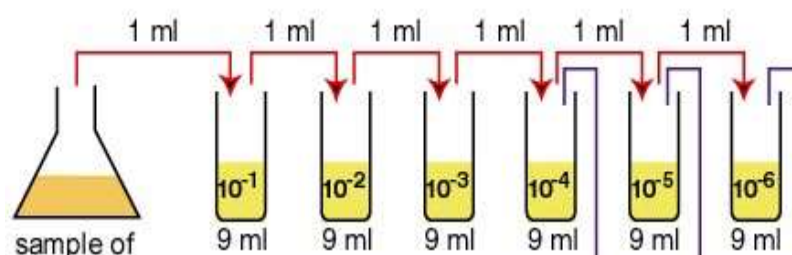
Step1 (day1):

11g of tomato(fresh weight) +99 ml St.D.water (sterile)

→ blending (2min low speed,2min high speed)

Homogenate sample

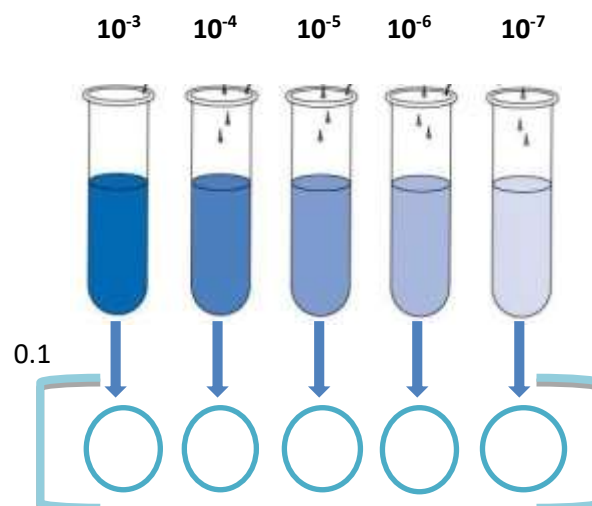
Step2: Dilution



9ml of H₂O

Step 3:Seeding

Each group inoculate 0.1 ml over NA(Nutrient agar) plate from each dilution



Surface spreading method
on Nutrient agar by L-shape
glass rod

Then, incubation 24hr/35c



L-Shape glass rod

Result:

- duplicate plate for each dilution
- be sure about number of colony use colony counter

Countable colonies:

Large colonies 3-20	
Medium colonies 3-300	pinpoint 3-1000

★ $CFU = \text{\# of colony} \times DF / \text{inoculation}$ ** $DF = 1/10^{-2} = 10^2$

هادي القانون مهم مهم
مهم/بيجي عليه أسئلة

- check presence of coliform and fecal coliform

* enterobacterial

*MPN: determine# of tubes that show gas after incubation time

- ★ for each dilution including 10^{-2} → 10^{-7} take 3 test tubes of (Lauryl sulfate Tryptose)with durham tube →

Incubate 24h/35c

After incubat:

*check for gas production → reincubation 24h,35c

*check for gas production & measure MPN

★ $MPN = MPN \text{ index} \times DF$

*يلي مو فاهم شو صار اخر شي : هلا بعد 24 ساعة بنعمل check for gas وبنرجع نحطهم
بال incubation 24 كمان و بنعمل check for gas and measure MPN



LAB#3(Coliform &fecal coliform organism);

Coliform:

Belong to Enterobacteriaceae

Found in intestine of human & animals

Used a good indication of health hazard

Most are harmless

Mild heat treatment (kill bacteria):eg: milk pasteurization



Fecal coliform:

-sub group of coliform eg: E-coli

Microscopic characters	E-coli
Gram reaction	-ve
Shape	Coccobacilli
Arrangment	Single
Spore formation	Non-spore former

-MPN: method for coliform and this method for drinking water

-Determine the number of tubes that shows gas after incubation

-3 test:1. Presumptive test /2.Complete test /3.Confirmatory test

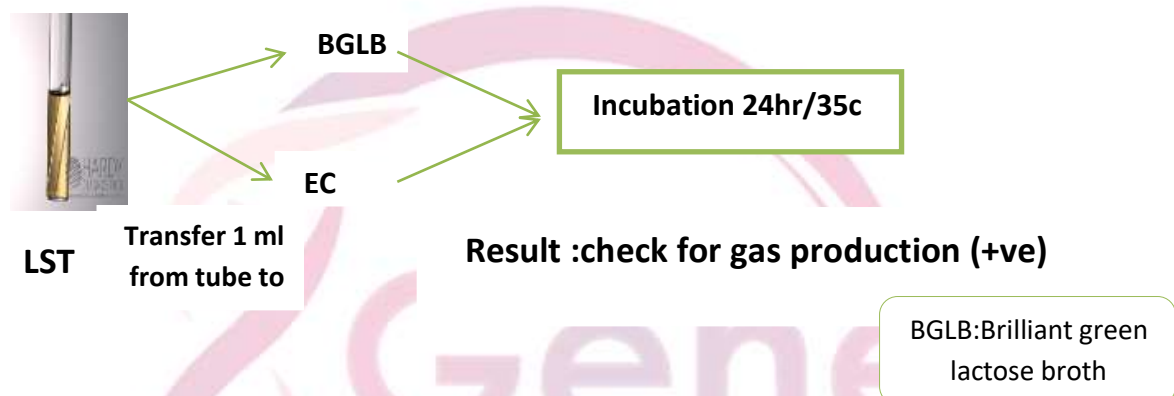
#LAB WORK



1. presumptive test(Day1,2):homogenous sample (tomato)(10^{-2}) → Incubation of LST(lauryl sulfate tryptose)

Result: how many test tube have gas?

2. confirmatory test(Day 3):



3.complete test (Day4):

Take one +ve of BGLB or EC → streaking quadrant method

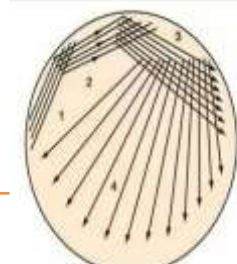
On EMB(Eosin Methylene Blue) → incubation 24hr/35c

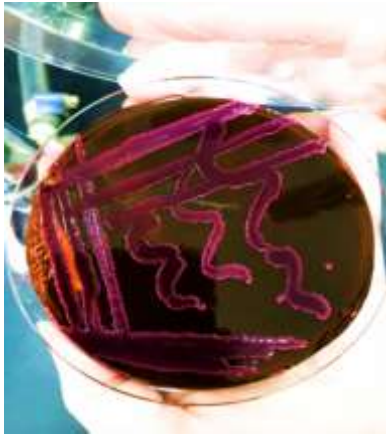
Result:

→ Klebsillea pneumonia on EMB agar: pink, mucoid glesting colonies

→ E.coli :Dark purple with black center and green metallic sheen

هادي هي طريقة زراعة البكتيريا بال
streaking quadrant وهادي
أضمن وأفضل طريقة الننا كطلاب





Klebsiella on EMB



E. coli on EMB

Day(5):

From EMB agar inoculate: (1.NA plate /2.NA slant) →
incubation 24hr/35c

Day(6):

From NA slant

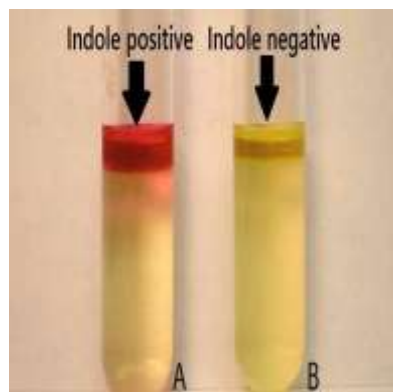
1.Gram staining: gram negative

:*coccobacilli,*nonsporeformer,*single arrangement

2.IMVIC test:

a.SIM(Sulfate indol motility) → take result after 24hr
(add Kovacs reagent)

Result: upper layer turns red (+ve) / layer turns yellow(-ve)



b.MR-VP:Methyl Red Vogas Proskaur

MR:add methyl red reagent /VP:add alpha-naphthol+40%KOH

→ **take result after 48hr**

Result:

MR: whole media red(that mean MR+ve)

Medium no change to red (MR-ve)

VP:red ring(that mean VP+ve)

No color change (VP-ve)

MR



VP



c.simmons citrate :take result after 48hr

Result:

The medium remains green (this is negative result)

The medium turns blue color or growth

(this is a positive result)



3.make pure culture



LAB#4 (Isolation of salmonella)

Microscopic characters	
Gram reaction	-ve
Shape	Coccobacilli
Arrangement	Single
Spore formation	Non-spore former
Family	Enterobacteriaceae

★-cause intestinal disease with vomiting and diarrhea

vomiting : تقيؤ

★-Isolate salmonella from food ,eg: chicken skin, eggs

diarrhea: اسهال

★-for salmonella six group include more than 1000 serovors which identify according to:

1. somatic(O) antigen
2. Flagellar(H) antigen

★ we have two methods to isolate salmonella:

→ A. ELISA –method : eg: API-system or Enterotube

→ B. Traditional-method:

ركزوا هون كثير مهمات

- 1.pre enrichment media :allow all types of microbes to grow

Eg:lactose broth

- 2. selective enrichment media :allow salmonella to grow suppress other

Eg: selenite broth

- 3.selective differential media : a.XLD Plate (xylose lysine disoxycholate agar) b.BS media (bismuth sulfite agar)

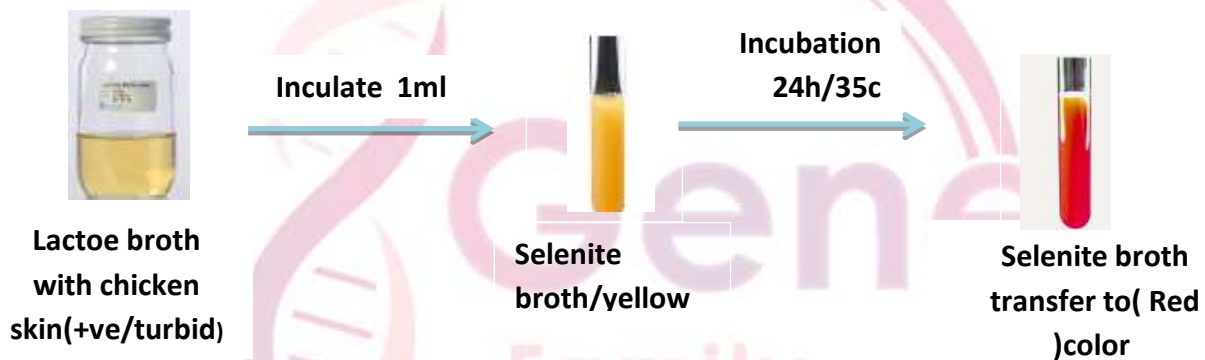
#LAB WORK:



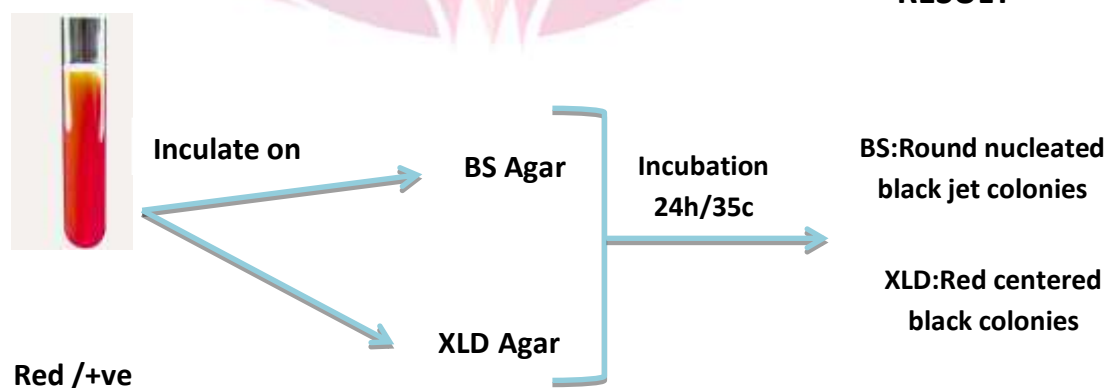
Day(1):

Take 10 g of (chicken skin) → lactose broth → shake
by hand (10 min)(vigorously) → incubation 24h/35c →
result → turbidity

Day (2):



Day(3):





**Salmonella colonies on
XLD Agar**



**Salmonella colonies on
BS Agar**

Questions:

1. The indication of bacterial growth on selenite broth is:

- a. Gas production b. Turbidity
c. Conversion to red d. Black color formation
-

2. Which of the following is a characteristic of Salmonella bacteria :

- a. Cocci b. Gram positive c. Spore former d. Single arrangement
-

3. The inhibitory agent included in the formula of XLD medium is:

- a. Xylose b. Lactose c. Sodium Deoxycholate d. Brilliant green
-

4. The growth of Staphylococcus aureus on XLD will appear as:

- a. Red colonies b. yellow colonies c. not grow
d. green colonies with metallic sheen

1.c	2.d	3.c	4.c
-----	-----	-----	-----

LAB#5 (ISOLATION OF STAPHYLOCOCCUS AUREUS)

Microscopic characters	
Gram reaction	+ve
Shape	Cocci
Arrangement	Grape like cluster
Spore formation	Non-spore former

-Isolation from food sample: falafel, meat,etc

-virulence factors:

→ **Toxins:** Enterotoxins (A, B, C, D, E)

Enterotoxins: proteins released by microorganism (causing food poisoning)

→ **Enzymes:**

1. catalase

2. coagulase (make clot in blood plasma)

3. nuclease (lysis the DNA)

→ **Pigments:** (staphyloxanthin)

Staphyloxanthin: pigment that produced by some strains of *S. aureus*

(وهادي الصبغة مسؤولة عن اللون الاصفر لـ *S. aureus*)

-sensitive to lysostaphin (is an extracellular enzyme secreted by strain of *S. simulans* biovar *staphylolyticus* with a bacteriocidal activity against staphylococci based on its capability of cleaving the peptidoglycan present in the bacterial cell walls)

#LAB WORK :

Day(1)

a.Blending

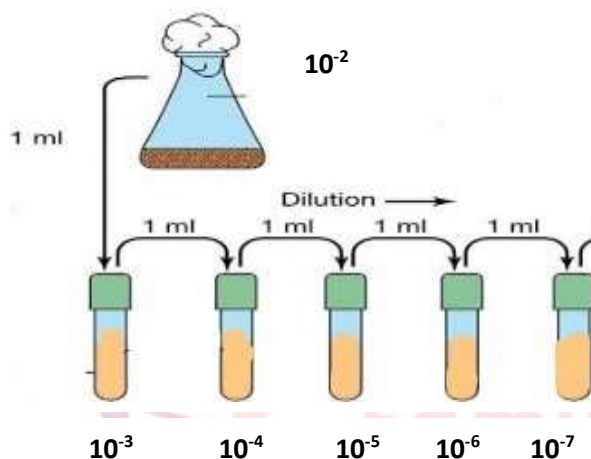
11g of falafel +99 ml of D.St.water)

Blending (2min low speed ,2min high speed)

homogenate sample 10^{-2}



b.Dilution: 10^{-2} to 10^{-7}



c.Seeding:

MSA

surface spreading method (L-shape glass rod) →

incubation 48hr

Result:

staphylococcus aureus (yellow colonies with yellow zone around the colony)

staphylococcus epidermidis (red cplonies with red zone)

E-coli (inhibited)

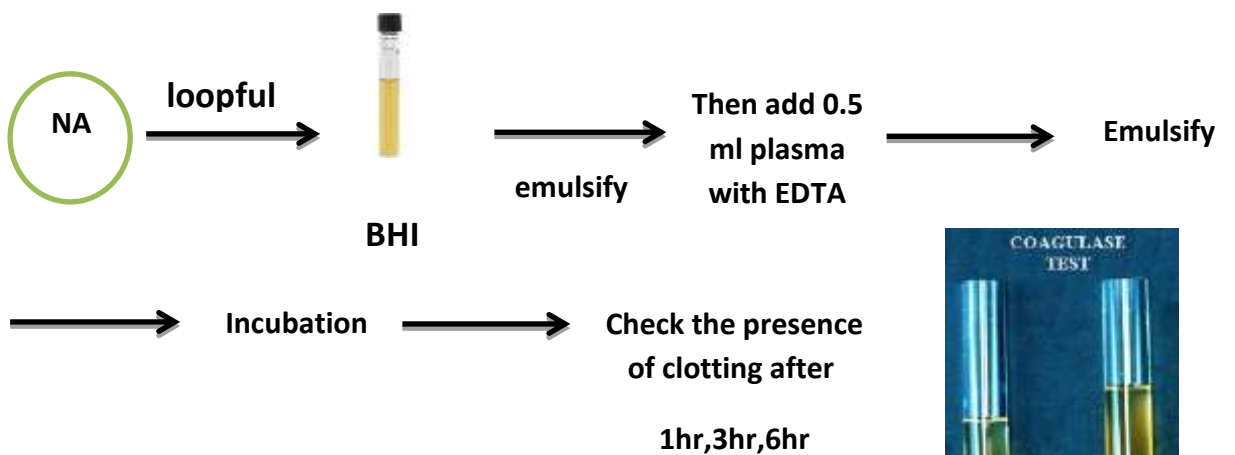
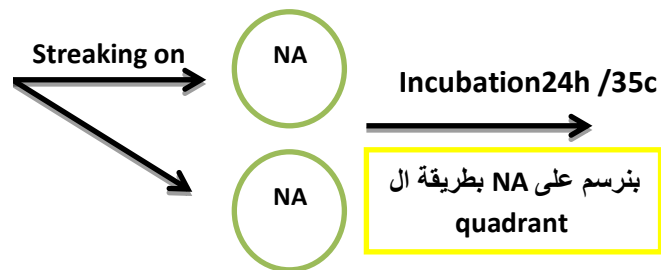


S.aureus

S.epidermidis

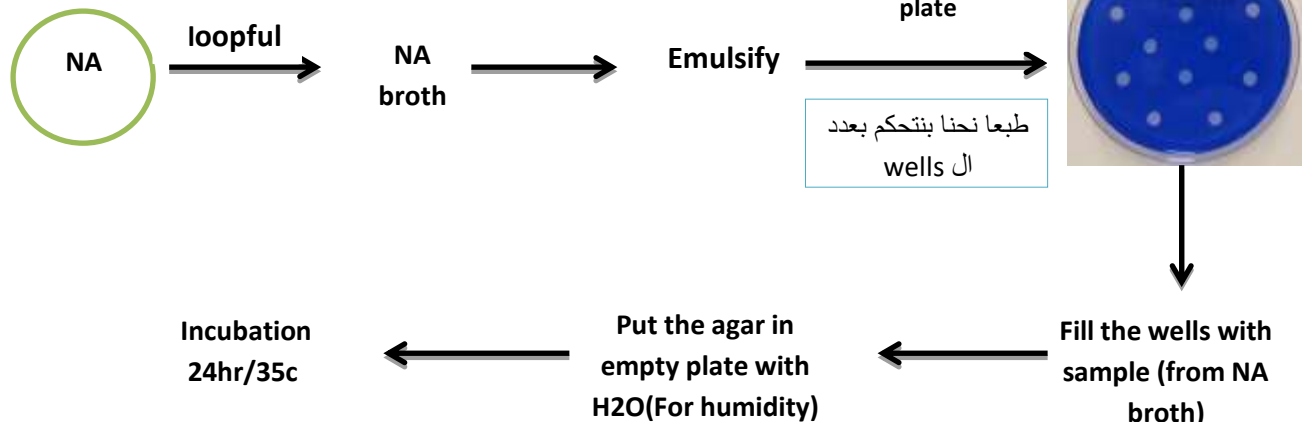


MSA

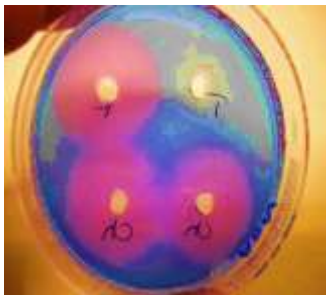


لاحظوا بالصورة انه positive يعني
100% S.aureus

*Nuclease test(Toludine Blue DNase Agar):



Result:



Red color around colonies:positive result/nuclease +ve

Still blue or other color:negative result/nuclease –ve

انتبهوا للصورة لتتوضح الفكرة أكثر

Family

Questions:

1.what is(are)the virulence factor(s)of staphylococcus aureus:

a.enzymes b.entotoxins c.pigments d. all of them

2.The ph indicator in Mannitol Salt Agar medium is:

**a.neutral red b.phenol red c.malachite green
d.Bromothymol blue**

3.all of the following is true about Staphylococcus epidermidis except:

**a.DNase +ve b.Coagulase-ve c.Catalase +ve
d.Lysostaphin sensitive**

Q.2:complete the following table with the missing word about the identification of Staphylococcus aureus:

Test	Media	Growth of S.aureus
	MSA	Yellow colonie with yellow zones
Coagulase		
	Toluidine blue DNA agar	Red color around the colonies

1.d 2.b 3.a

LAB#6(ISOLATION OF BACILLUS CEREUS)

Microscopic characters	
Gram reaction	+ve
Shape	Rods(square shape)
Arrangmaent	Chains
Spore formation	Central/subcentral without swelling

★ found in soil,dust,air,vegetables,fresh and salt water sources

★ Pathogenicity:

Self-limited gastroenteritis of two type:1.Emetic 2.Diarrheal



Criteria \ Type	Emetic	Diarrheal
Enzyme(s)	Cereulide	1. Hemolysin BL (HBL) 2. Nonhemolytic Enterotoxin (NHE) 3. Cytotoxin K
Heat lability	a heat- and pH-stable	heat labile enterotoxins
Incubation period	1 to 5 hours	10 to 12 hours
Symptoms	Nausea and vomiting	Abdominal pain with diarrhea
Similar Enterotoxins produced by	<i>Staphylococcus aureus</i>	<i>Clostridium perfringens</i>
Example food	Starchy food e.g. fried rice	Proteinaceous food e.g. meat

#LAB WORK :

Day (1):



★ step(1):

1g of soil +99 ml of St.D.H2O $\xrightarrow{\text{Shake by hand}}$ homogenate sample(10^{-2})

★ Step(2):

Dilution: $10^{-2} \rightarrow 10^{-5}$

★ Step(3):

Seeding

For each group 2 plate of (phenol red egg yolk polymyxin agar)(MYP) $\rightarrow$ Incubation 48hr/35c

MYP:selective media for bacillus cereus



Result:

Bacillus cereus:red colonies with opaque zone around the colony (mannitol non fermenter)-ve



Bacillus subtilis:yellow colonie without zone around the colony (monnitol fermenter)+ve



Day(2):

From MYP culture $\xrightarrow{\text{streaking}}$ Nutrient agar $\xrightarrow{\text{incubation 24hr}}$

Make the following test of bacillus cereus :

1.phenol red glucose broth with Derhum tube:

Nutrient agar $\xrightarrow{\text{loopful}}$  $\xrightarrow{\text{Mix very well, then incubation 48 hr/35c}}$

Result: medium turns yellow with no gas (+ve) bacillus cereus 100%

يعني ال phenol red glucose بيكون أحمر وبيتحول للون الأصفر ويننتبه انه ما عندي gas in derhum tube وهاه ببدل انه نسبة ال bacillus cereus 100%



Note:

Phenol red :red in nutrient /yellow in acid /pink in alkaline



2. Tyrosine agar slant:

Tyrosine: source of amino acid

NA



loopful



Incubation 48hr/35c

Result:

clear zone around the bacterial growth (colorless area) +ve reaction / that mean bacillus cereus able to hydrolyze tyrosine.

3. MR-VP: methyl-red vogas proskauer test:

NA



MR-VP broth



incubation 48hr

Result:

1. add methyl red → whole media red (glucose stable), (that mean MR +ve)

The medium shows no change to red
(that mean MR -ve)



2. add Barritts A reagent : consist of alpha-naphthol + 40% KOH

→ red ring (glucose unstable acid) / (that mean VP +ve)

NO color change (VP -ve)

Red
Ring



4. Growth with lysozyme (0.001%)

1. control → from pure culture, inoculate loopful in NB →
mix → incubation 24hr/35c

Result: Turbidity

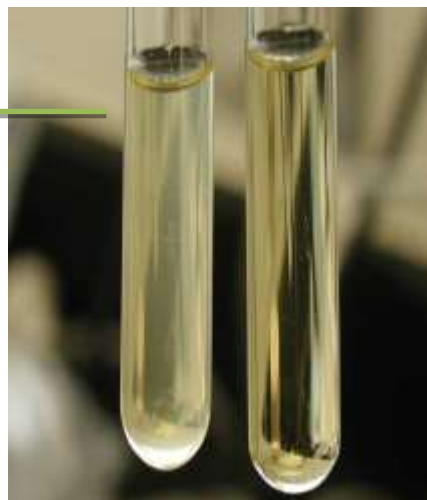
2. treatment → inoculate loopful in NB → mix
add lysozyme (0.5ml) → emulsify → incubation
48hr/35c

Result :

*Turbidity: bacillus cereus (can live in lysozyme)

*not turbid : no bacillus cereus

Turbidity



Questions:

1.You isolate bacillus cereus from:

- a.tomato b. milk c. chicken skin d. soil**
-

2.Bacillus cereus produces a type of thermostable toxins called emetic toxins: a.True b.false

3.which of the following is a characteristic of bacillus cereus bacteria :

- a.Grape-like clusters b.Bacilli c.Gram negative**
-

4. on Mannitol Yolk Polymyxin Agar,the appearance of yellow colonies with yellow zones around the colonies is an indication for:

- a.lecithinase activity b.Polymyxin B sensitivity**
c.mannitol fermentation d. none of them

1.d	2.a	3.b	4.c
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